

POTHOLELOG · DESIGN DECISION RECORD

PotholeLog recommendations accepted

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DOCUMENT CONTROL

Document ID	PHL-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potholelog		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 (first pilot partner) and the CityTwin ingest path decided by Amish on 2026-10-02
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Items D1 to D7, N1 and N2 are "Decided by Amish, 2026-09-25: go with recommendation". Item O1 was decided by Amish on 2026-10-02 (PHL-DEC-001).

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Every item in `docs/REVIEW.md` and PHL-DDR-001 that was awaiting him and carried a recommendation is therefore decided as recommended. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's instruction, so nothing here authorizes building, testing or purchasing.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Bicycles in the pitch	Keep the pitch; a bicycle variant (own battery, handlebar or seat-post mount) is a later step	Status wording only; R16 already served buses and refuse trucks; pitch unchanged
D2	Upload route	Depot Wi-Fi first; LTE-M cellular as an optional add-on, costed separately	Status wording only
D3	Mount location	Sprung-mass floor mount above the rear axle	Status wording only
D4	Controller	ESP32-S3 board with Wi-Fi and microSD slot	Status wording in PHL-PRC-001 and bom/bom.csv
D5	IMU and GNSS classes	LSM6DSO-class IMU and u-blox M10-class GNSS, to confirm against datasheets	Status wording in bom/bom.csv
D6	Segment length and output format	100 m segments plus point events, in CSV and GeoJSON following the CityTwin open data export	Status wording only; CityTwin ingest path listed as a cross-repo action
D7	Privacy rule	Publish only road-segment roughness and defect clusters; tracks and timestamps stay with the operator; no driver scoring	Status wording in PHL-PRB-001
N1	Converter input for 24 V vehicles (R9)	Option (a): a 60 V-rated converter, with budget_usd raised to \$75	bom/bom.csv item 4 from 9 to 36 V at \$7.00 to 9 to 60 V at \$9.00; parts cost \$69.00 to \$71.00; budget_usd \$70 to \$75 (margin \$1.00 to \$4.00); R15 target \$70 to \$75; R9 from not met to at risk (1.9 V margin over the TVS clamp, TVS pulse energy unverified); PHL-DWG-001 supply note, Rev P1 to P2; blueprint key figures; PHL-CAL-001 F4, F5, J1, J2
N2	Defect-detection target (R1, R5)	Option (b): limit the R5 defect-detection range to 10 to 30 km/h and rely on repeat slow passes near stops and junctions, keeping the 300 mm reference defect	R5 target restated in PHL-REQ-001 (was 10 to 80 km/h); R5 from not met to at risk (lowest margin 1.94 on IRI 2 and 0.97 at 30 km/h on IRI 4; new line K2 in PHL-CAL-001); PHL-PRC-001 and PHL-PRB-001 updated

No design change altered geometry: the 60 V converter keeps the 40 x 30 x 14 mm envelope in cad/src/model.py. STEP, STL, the drawing and the concept media were regenerated.

Table 2. Items still open.

#	Item	Status
O1	First pilot partner and city, and who provides reference IRI	No recommendation was made then. Decided by Amish, 2026-10-02 (PHL-DEC-001): first candidate to approach a Dallas-Fort Worth area city with its own bus or refuse fleet, with TxDOT pavement management roughness data on state roads in that city as the reference IRI sections

■ Cross-repo actions

- **CityTwin ingest path (D6, R14).** CityTwin's gateway accepts no inbound connections, and PotholeLog uploads to the fleet operator's server. The two repos must agree whether CityTwin fetches the operator's daily segment files or the operator passes them on by another route, and confirm that a daily publication date is compatible with R13. Recorded here; CityTwin is not edited. Decided 2026-10-02 (PHL-DEC-001): CityTwin fetches the daily segment files from the operator's server (option a, matching CityTwin's outbound pull, CTW-DDR-001 D11); a date per segment may be published but never pass times, and only once at least two vehicles or several days are pooled.

■ Consequences

- Requirement status (PHL-CAL-001 v0.2): 0 not met, 6 at risk (R1, R4, R5, R9, R11, R12), 2 not verifiable at TRL 3 (R2, R3), 4 met on paper (R7, R8, R10, R15), 4 met by design (R6, R13, R14, R16). Before: 2 not met, 4 at risk.
- Decided but on hold because they are TRL 4 work: a bench test of the TVS and converter against ISO 16750-2 load dump pulses (N1), and field data on slow-pass detection rates (N2).
- `trl: 3` and `trl_target: 3` are unchanged.
- PHL-PRB-001, PHL-PRC-001 and PHL-REQ-001 are revised to v0.4; PHL-CAL-001 and PHL-DDR-001 to v0.2.